

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which statements about the `final` modifier are correct? (Choose all that apply.)
 - A. Instance and static variables can be marked `final`.
 - B. A variable is effectively final only if it is marked `final`.
 - C. An object that is marked `final` cannot be modified.
 - D. Local variables cannot be declared with type `var` and the `final` modifier.
 - E. A primitive that is marked `final` cannot be modified.
2. Which of the following can fill in the blank in this code to make it compile? (Choose all that apply.)

```
public class Ant {  
    ____ void method() {}  
}
```

- A. `default`
 - B. `final`
 - C. `private`
 - D. `Public`
 - E. `String`
 - F. `zzz:`
3. Which of the following methods compile? (Choose all that apply.)
 - A. `final static void rain() {}`
 - B. `public final int void snow() {}`
 - C. `private void int hail() {}`
 - D. `static final void sleet() {}`
 - E. `void final ice() {}`
 - F. `void public slush() {}`
 4. Which of the following can fill in the blank and allow the code to compile? (Choose all that apply.)

```
final _____ song = 6;
```

 - A. `int`
 - B. `Integer`
 - C. `long`
 - D. `Long`
 - E. `double`
 - F. `Double`

5. Which of the following methods compile? (Choose all that apply.)

- A. `public void january() { return; }`
- B. `public int february() { return null;}`
- C. `public void march() {}`
- D. `public int april() { return 9;}`
- E. `public int may() { return 9.0;}`
- F. `public int june() { return;}`

6. Which of the following methods compile? (Choose all that apply.)

- A. `public void violin(int... nums) {}`
- B. `public void viola(String values, int... nums) {}`
- C. `public void cello(int... nums, String values) {}`
- D. `public void bass(String... values, int... nums) {}`
- E. `public void flute(String[] values, ...int nums) {}`
- F. `public void oboe(String[] values, int[] nums) {}`

7. Given the following method, which of the method calls return 2? (Choose all that apply.)

```
public int juggle(boolean b, boolean... b2) {  
    return b2.length;  
}
```

- A. `juggle();`
- B. `juggle(true);`
- C. `juggle(true, true);`
- D. `juggle(true, true, true);`
- E. `juggle(true, {true, true});`
- F. `juggle(true, new boolean[2]);`

8. Which of the following statements is correct?

- A. Package access is more lenient than protected access.
- B. A public class that has private fields and package methods is not visible to classes outside the package.
- C. You can use access modifiers so only some of the classes in a package see a particular package class.
- D. You can use access modifiers to allow access to all methods and not any instance variables.
- E. You can use access modifiers to restrict access to all classes that begin with the word Test.

9. Given the following class definitions, which lines in the `main()` method generate a compiler error? (Choose all that apply.)

// Classroom.java

```
package my.school;

public class Classroom {
    private int roomNumber;
    protected static String teacherName;
    static int globalKey = 54321;
    public static int floor = 3;
    Classroom(int r, String t) {
        roomNumber = r;
        teacherName = t; } }
```

// School.java

```
1: package my.city;
2: import my.school.*;
3: public class School {
4:     public static void main(String[] args) {
5:         System.out.println(Classroom.globalKey);
6:         Classroom room = new Classroom(101, "Mrs. Anderson");
7:         System.out.println(room.roomNumber);
8:         System.out.println(Classroom.floor);
9:         System.out.println(Classroom.teacherName); } }
```

A. None: the code compiles fine.

B. Line 5

C. Line 6

D. Line 7

E. Line 8

F. Line 9

10. What is the output of executing the Chimp program?

// Rope.java

```
1: package rope;
2: public class Rope {
3:     public static int LENGTH = 5;
4:     static {
5:         LENGTH = 10;
6:     }
7:     public static void swing() {
```

```
8:      System.out.print("swing ");
9:  } }
```

// Chimp.java

```
1: import rope.*;
2: import static rope.Rope.*;
3: public class Chimp {
4:     public static void main(String[] args) {
5:         Rope.swing();
6:         new Rope().swing();
7:         System.out.println(LENGTH);
8:     } }
```

- A. swing swing 5
 - B. swing swing 10
 - C. Compiler error on line 2 of Chimp
 - D. Compiler error on line 5 of Chimp
 - E. Compiler error on line 6 of Chimp
 - F. Compiler error on line 7 of Chimp
11. Which statements are true of the following code? (Choose all that apply.)

```
1: public class Rope {
2:     public static void swing() {
3:         System.out.print("swing");
4:     }
5:     public void climb() {
6:         System.out.println("climb");
7:     }
8:     public static void play() {
9:         swing();
10:        climb();
11:    }
12:    public static void main(String[] args) {
13:        Rope rope = new Rope();
14:        rope.play();
15:        Rope rope2 = null;
16:        System.out.print("-");
17:        rope2.play();
18:    } }
```

- A. The code compiles as is.
- B. There is exactly one compiler error in the code.
- C. There are exactly two compiler errors in the code.
- D. If the line(s) with compiler errors are removed, the output is swing-climb.
- E. If the line(s) with compiler errors are removed, the output is swing-swing.
- F. If the line(s) with compile errors are removed, the code throws a NullPointerException.

12. How many variables in the following method are effectively final?

```
10: public void feed() {  
11:     int monkey = 0;  
12:     if(monkey > 0) {  
13:         var giraffe = monkey++;  
14:         String name;  
15:         name = "geoffrey";  
16:     }  
17:     String name = "milly";  
18:     var food = 10;  
19:     while(monkey <= 10) {  
20:         food = 0;  
21:     }  
22:     name = null;  
23: }
```

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. None of the above. The code does not compile.

13. What is the output of the following code?

```
// RopeSwing.java  
import rope.*;  
import static rope.Rope.*;  
public class RopeSwing {  
    private static Rope rope1 = new Rope();  
    private static Rope rope2 = new Rope();  
    {  
        System.out.println(rope1.length);  
    }  
}
```

```
public static void main(String[] args) {  
    rope1.length = 2;  
    rope2.length = 8;  
    System.out.println(rope1.length);  
}  
}
```

// Rope.java

```
package rope;  
public class Rope {  
    public static int length = 0;  
}
```

- A. 02
- B. 08
- C. 2
- D. 8**
- E. The code does not compile.
- F. An exception is thrown.

14. How many lines in the following code have compiler errors?

```
1: public class RopeSwing {  
2:     private static final String leftRope;  
3:     private static final String rightRope;  
4:     private static final String bench;  
5:     private static final String name = "name";  
6:     static {  
7:         leftRope = "left";  
8:         rightRope = "right";  
9:     }  
10:    static {  
11:        name = "name";  
12:        rightRope = "right";  
13:    }  
14:    public static void main(String[] args) {  
15:        bench = "bench";  
16:    }  
17: }
```

- A. 0
- B. 1

- C. 2
- D. 3
- E. 4
- F. 5

15. Which of the following can replace line 2 to make this code compile? (Choose all that apply.)

```
1: import java.util.*;
2: // INSERT CODE HERE
3: public class Imports {
4:     public void method(ArrayList<String> list) {
5:         sort(list);
6:     }
7: }
```

- A. `import static java.util.Collections;`
- B. `import static java.util.Collections.*;`
- C. `import static java.util.Collections.sort(ArrayList<String>);`
- D. `static import java.util.Collections;`
- E. `static import java.util.Collections.*;`
- F. `static import java.util.Collections.sort(ArrayList<String>);`

16. What is the result of the following statements?

```
1: public class Test {
2:     public void print(byte x) {
3:         System.out.print("byte-");
4:     }
5:     public void print(int x) {
6:         System.out.print("int-");
7:     }
8:     public void print(float x) {
9:         System.out.print("float-");
10:    }
11:    public void print(Object x) {
12:        System.out.print("Object-");
13:    }
14:    public static void main(String[] args) {
15:        Test t = new Test();
16:        short s = 123;
17:        t.print(s);
18:        t.print(true);
19:    }
20: }
```

```

19:         t.print(6.789);
20:     }
21: }

```

- A. byte-float-Object-
- B. int-float-Object-
- C. byte-Object-float-
- D. int-Object-float-
- E. int-Object-Object-**
- F. byte-Object-Object-

17. What is the result of the following program?

```

1: public class Squares {
2:     public static long square(int x) {
3:         var y = x * (long) x;
4:         x = -1;
5:         return y;
6:     }
7:     public static void main(String[] args) {
8:         var value = 9;
9:         var result = square(value);
10:        System.out.println(value);
11:    } }

```

- A. -1
- B. 9**
- C. 81
- D. Compiler error on line 9
- E. Compiler error on a different line

18. Which of the following are output by the following code? (Choose all that apply.)

```

public class StringBuilders {
    public static StringBuilder work(StringBuilder a,
        StringBuilder b) {
        a = new StringBuilder("a");
        b.append("b");
        return a;
    }
    public static void main(String[] args) {
        var s1 = new StringBuilder("s1");
        var s2 = new StringBuilder("s2");

```

```

        var s3 = work(s1, s2);
        System.out.println("s1 = " + s1);
        System.out.println("s2 = " + s2);
        System.out.println("s3 = " + s3);
    }
}

```

- A. s1 = a
- B. s1 = s1
- C. s2 = s2
- D. s2 = s2b
- E. s3 = a
- F. The code does not compile.

19. Which of the following will compile when independently inserted in the following code? (Choose all that apply.)

```

1: public class Order3 {
2:     final String value1 = "red";
3:     static String value2 = "blue";
4:     String value3 = "yellow";
5:     {
6:         // CODE SNIPPET 1
7:     }
8:     static {
9:         // CODE SNIPPET 2
10:    } }

```

- A. Insert at line 6: value1 = "green";
- B. Insert at line 6: value2 = "purple";
- C. Insert at line 6: value3 = "orange";
- D. Insert at line 9: value1 = "magenta";
- E. Insert at line 9: value2 = "cyan";
- F. Insert at line 9: value3 = "turquoise";

20. Which of the following are true about the following code? (Choose all that apply.)

```

public class Run {
    static void execute() {
        System.out.print("1-");
    }
    static void execute(int num) {
        System.out.print("2-");
    }
}

```

```

static void execute(Integer num) {
    System.out.print("3-");
}
static void execute(Object num) {
    System.out.print("4-");
}
static void execute(int... nums) {
    System.out.print("5-");
}
public static void main(String[] args) {
    Run.execute(100);
    Run.execute(100L);
}
}

```

- A. The code prints out 2-4-.
 - B. The code prints out 3-4-.
 - C. The code prints out 4-2-.
 - D. The code prints out 4-4-.
 - E. The code prints 3-4- if you remove the method `static void execute(int num)`.
 - F. The code prints 4-4- if you remove the method `static void execute(int num)`.
21. Which method signatures are valid overloads of the following method signature? (Choose all that apply.)
- ```
public void moo(int m, int... n)
```
- A. `public void moo(int a, int... b)`
  - B. `public int moo(char ch)`
  - C. `public void moooo(int... z)`
  - D. `private void moo(int... x)`
  - E. `public void moooo(int y)`
  - F. `public void moo(int... c, int d)`
  - G. `public void moo(int... i, int j...)`